



ZIMBABWE INTEGRATING CAPITAL AND RISK PROGRAMME

TS 6.2 - Technical Specification for the Determination of Life Underwriting Risk Capital Requirement

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1. Introduction

- 1.1 Life Underwriting Risk ('LUR') is the risk arising from life insurance obligations, such as poor claims experience, expense overruns and policy lapses. It comprises the following components:
- i. Mortality risk;
 - ii. Longevity risk;
 - iii. Disability/morbidity risk;
 - iv. Life expense risk;
 - v. Life catastrophe risk;
 - vi. Retrenchment risk; and
 - vii. Lapse risk.
- 1.2 The Zimbabwe Integrating Capital and Risk Programme ('ZICARP') framework makes use of a correlation matrix for the aggregation of the above sub risk modules.
- 1.3 The following safeguards are considered important in selecting correlation parameters under the ZICARP framework:
- i. To take note of any dependencies that would not be addressed by linear correlations;
 - ii. To choose the correlation coefficients to adequately reflect potential dependencies in the distribution tails; and
 - iii. To assess the stability of any correlation assumptions under stress conditions, that is to conduct a sensitivity analysis.

2. Overview of Life Underwriting Risk

- 2.1 The Solvency Capital Requirement ('SCR') standard formula under ZICARP follows a modular approach. The overall risk which the insurance or reinsurance undertaking is exposed to is divided into sub-risk modules.
- 2.2 For each sub-risk module, a capital requirement is determined. The capital requirements on sub-risk level are aggregated in order to derive the capital requirement for the overall risk.

- 2.3 The capital requirement for life underwriters, SCR_{Life} shall be calculated as:

$$SCR_{Life} = \sum_i \sum_j CorrLU_{ij} \cdot LU_i \cdot LU_j$$

Where:

LU_i = The life underwriting sub-module i ;

LU_j = The life underwriting sub-module j ; and

$CorrLU_{ij}$ = The correlation parameter between components i and j .

- 2.4 The capital requirement for each sub-module has been calibrated to correspond to a 99.5% Value-at-Risk.

- 2.5 In the actual calculation, LU_i and LU_j will be replaced by:

LU_{mort} which denotes the mortality risk sub-module;

LU_{long} which denotes the longevity risk sub-module;

LU_{dis} which denotes the disability/morbidity risk sub-module;

LU_{lapse} which denotes the lapse risk sub-module;

LU_{exp} which denotes the life expense risk sub-module;

LU_{CAT} which denotes the life catastrophe risk sub-module; and

LU_{ret} which denotes the retrenchment risk sub-module.

- 2.6 The correlation matrix $CorrLU$ is defined as:



<i>CorrLU</i>	Mortality	Longevity	Disability	Lapse	Expense	Catastrophe	Retrenchment
Mortality	1	-0.25	0.25	0	0.25	0.25	0
Longevity		1	0	0.25	0.25	0	0
Disability			1	0	0.5	0.25	0
Lapse				1	0	0.25	0.25
Expense					1	0.25	0.25
Catastrophe						1	0
Retrenchment							1

Table 2.1: The correlation matrix, *CorrLU*

3. Capital Calculation for the Sub-Risk Modules

Mortality Risk

- 3.1 Mortality risk arises when there is an increase in mortality rates relative to the best estimate assumptions.
- 3.2 Its solvency capital requirement, LU_{mort} , per line of business shall be determined according to the formula;

$$LU_{mort} = 15\% \cdot q \cdot n \cdot 1.1^{\frac{n-1}{2}} \cdot \sum_i \max(Death\ Benefit_i - BEL_i, 0)$$

Where:

- i. The amount the insurer would currently pay in the event of the death of the persons insured under the policy; plus
- ii. The expected present values of amounts not covered in (i) the insurer would pay in the event of the immediate death of the persons insured under the policy;

BEL_i = The insurer-specific expected average death rate (including the best estimate assumption for HIV/AIDS) over the coming 12 months, weighted by the sum assured; and

n = The modified duration of the liability cash flows which must be at least one year.

- 3.3 The total mortality risk faced by the insurer should be determined as the summation of mortality risk capital requirement for all lines of business that are subject to mortality risk.
- 3.4 If there exist insurance obligations that provide benefits on both death and survival of the same insured individual, allowance may be made for the

netting effect provided by the hedge between death and survival benefits components.

3.5 The insurer may assume that management actions under stressful conditions, such as adjustments to future bonus rates on policies with discretionary participation features, may occur when determining the capital requirement for mortality risk.

3.6 Whether the stress is insurer-specific or industry-wide, must be taken into account when determining the type and scope of management actions the insurer may take.

Longevity risk

3.7 Longevity risk arises when there is a decrease in mortality rates relative to the best estimate assumptions.

3.8 The Longevity risk solvency capital requirement has two components:

- i. The annuity component, $Long_{Annuities}^{LU}$; and
- ii. The component for other policies subjected to longevity risk, $Long_{Other}$

3.9 The measure $Long_{Annuities}$ is given by the formula:

$$Long_{Annuities} = 13.8\% \times \sum_i \max(BEL_{iAnnuities_{long}}, 0) \times (1 - q) \times n \times 1.1^{\frac{n-1}{2}}$$

Where:

$BEL_{iAnnuities_{long}}$ = Best estimate liability for annuity policy i subject to longevity risk; and

q and n = The same measures as defined for mortality risk.

3.10 The measure $Long_{Other}$ is given by the formula:

$$Long_{Other} = 13.8\% \times q \times n \times 1.1^{\frac{n-1}{2}} \times \sum_i \max(Bell_{i_{Other_{long}}}, 0)$$

Where:

$Bell_{i_{Other_{long}}}$ = Best estimate liability for any other policy i that is subject to longevity risk; and

q and n = The same measures as defined for mortality risk.

3.11 LU_{long} is then the sum of the above two measures, i.e.,

$$LU_{Long} = Long_{Annuities} + Long_{Other}$$

3.12 If there exist insurance obligations that provide benefits on both death and survival of the same insured individual, allowance may be made for the netting effect provided by the hedge between death and survival benefits components.

3.13 The insurer may make the assumption that management actions under stress scenarios, such as adjustments to future bonus rates on policies with discretionary participation elements, may occur when determining the capital required for longevity risk.

3.14 Whether the stress is peculiar to the insurer or the entire industry must be taken into account when determining the kind and scope of management actions that the insurer may take.

Disability/morbidity risk

3.15 Disability/morbidity risk arises when there is an increase in the disability/morbidity inception rates or a decrease in disability/morbidity.

3.16 The capital requirement for disability risk, LU_{dis} , is calculated according to the formula;

$$LU_{dis} = 25\% \cdot \sum_i \max(Disability\ Benefit_i - BEL_i, 0) \cdot q \cdot n \cdot 1.1^{\frac{n-1}{2}} + 20\% \cdot BEL_t \cdot t \cdot n \cdot 1.1^{\frac{n-1}{2}}$$

where:

Disability Benefit_i =

- i. The amount the insurer would currently pay should the persons insured under the policy experience disability or morbidity; plus
- ii. The expected present values of amounts not covered in (i) the insurer would pay should there be immediate disability or morbidity of the persons insured under the policy;

BEL_i = The best estimate liabilities of the corresponding obligations in points (i) and (ii);

q = The insurer-specific expected average rate of transition from healthy to sick or disabled over the coming 12 months, weighted by the sum assured or annual payment;

n = The modified duration of the liability cash-flows;

BEL_t = The best estimate liability for insurance obligations subject to termination or recovery risk; and

t = The insurer-specific average rate of transition from sick or disabled to healthy or dead over the coming 12 months, weighted by the sum assured or annual payment.

3.17 In calculating the capital requirement for disability/morbidity risk, the insurer may assume that management actions under stress scenarios can take place including changes to future bonus rates on policies with discretionary participation features.

3.18 In calculating the capital requirement for disability/morbidity risk, the insurer may assume that management actions under stress scenarios can take place including changes to future bonus rates on policies with discretionary participation features.

Lapse risk

3.19 Lapse risk arises when there is an increase in lapse rates either for new or in-force business.

3.20 The capital requirement for lapse risk, LU_{lapse} , is calculated by the formula:

$$LU_{lapse} = \max(Lapse_{down}, Lapse_{mass}, Lapse_{up})$$

where:

$Lapse_{down}$ = Capital requirement for the risk of a permanent decrease of the rates of lapsation;

$Lapse_{up}$ = Capital requirement for the risk of a permanent increase of the rates of lapsation; and

$Lapse_{mass}$ = The capital requirement for lapse mass event.

3.21 The measures $Lapse_{up}$ and $Lapse_{down}$ are calculated as:

$$Lapse_{up} = 50\% \times l_+ \times n_+ \times S_+$$

$$Lapse_{down} = 50\% \times l_- \times n_- \times S_-$$

where:

l_+ = Estimate of the average rate of option exercise of the homogeneous groups with a positive surrender strain;

n_+ = The average number of years (weighted by surrender strain) which the homogeneous groups with a positive surrender strain runoff, subject to a minimum of one year;

S_+ = The sum of positive surrender strains;

l_- = Estimate of the average rate of option exercise of the homogeneous groups with a negative surrender strain;

n_- = The average number of years (weighted by surrender strain) over which the homogeneous groups with a negative surrender strain runoff, subject to a minimum of one year; and S_- = The sum of negative surrender strains.

3.22 The measure $Lapse_{mass}$ is calculated as:

$$Lapse_{mass} = \alpha \times S_+$$

where $\alpha = 30\%$ and for non-retail business, put $\alpha = 70\%$.

3.23 In calculating the capital requirement for lapse risk, an insurer may assume that management actions can take place under stress scenarios, including changes to future bonus on policies with discretionary participation features.

Expense risk

3.24 Expense risk arises when there is an increase in expenses or inflation relative to the best estimate assumptions.

3.25 The capital requirement for expense risk, LU_{exp} , is given by the formula:

$$LU_{exp} = (10\% \times n + (S_{n|@k\%} - S_{n|@i\%})) \times pE$$

where:

n = The modified duration of the liability cash flows associated with life insurance obligations serviced during the previous 12 months, which must be set to not less than one year;

pE = The amount of expenses incurred in servicing life insurance obligations during the previous 12 months;

k = The stressed inflation rate calculated as:

An increase in future expenses relative to the best estimate assumptions; plus

{an absolute addition of to the best estimate level of expense inflation, an increase in the best estimate level of expense inflation}; and

i = Weighted average inflation rate included in the calculation of the best estimate of life insurance obligations serviced during the previous 12 months, weighted by the present value of expenses included in the calculation of the best estimate for servicing all existing life insurance obligations.

3.26 The insurer may assume that management actions under stress scenarios may occur, including alterations to future bonus rates on plans with discretionary participation features, when determining the capital requirement for expense risk.

- 3.27 Insurers should only consider reasonable management activities in connection to changeable expenditure loadings for policies. Whether the stress is peculiar to the insurer or the entire industry must be taken into account when determining the kind and scope of management actions that the insurer may take.

Catastrophe risk

- 3.28 Catastrophe risk for life insurance obligations is the risk of loss, or of adverse change in the value of insurance obligations, resulting from extreme or irregular events whose effects are not sufficiently captured by the other risk components of life underwriting risk. Examples include pandemic events (e.g. Covid-19, HIV and AIDS, etc.) and disasters such as nuclear disasters.

- 3.29 The capital requirement for life catastrophe risk, LU_{CAT} , is calculated by the formula:

$$LU_{CAT} = \sqrt{\sum_{i,j} CorrLU_{i,j} \cdot CAT_i \cdot CAT_j}$$

where:

i, j = Mortality and morbidity sub-modules respectively; and

$CorrLU$ = The correlation matrix given below:

$CorrCATLU$	Mortality	Morbidity
Mortality	1	0.25
Morbidity	0.25	1

Table 2.2: The correlation matrix $CorrCATLU$

3.30 The measure CAT_{mort} is calculated by the formula:

$$CAT_{mort} = \sum_i MortCAT\ shock_{rate} \times CAR_i$$

Where:

i = Each policy where the payment of benefits is contingent upon mortality.

$$CAR_i = SA_i + SA_i \cdot Annuity_factor - BEL_i$$

SA_i = The sum assured (net of reinsurance) on death for each policy i where the benefits are payable as a single lump sum;

BA_i = The annualised amount of benefit (net of reinsurance) payable on death for each policy i where the benefits are not payable as a single lump sum;

$Annuity_factor$ = The average annuity factor for the expected duration over which benefits may be payable in the event of a claim; and

BEL_i = The best estimate liability (net of reinsurance) for each policy i .

$$MortCAT\ shock_{rate} = \frac{12 \times \min\{\max(0.2 \times MortRate + 0.105, 0.125), 0.3\}}{1000}$$

$MortRate$ = The underlying mortality rate per thousand per month per life insured.

3.31 The measure CAT_{morb} is calculated according to the formula:

$$CAT_{morb} = \sum_j MorbCAT\ shock_{rate} \times CAR_j$$

where:

j = Each policy where the payment of benefits is contingent on disability or sickness;

$$MorbCAT\ shock_{rate} = \frac{12 \times 70\% \times MorbRate}{1000};$$

$$CAR_j = SA_j + BA_j \times Annuity_factor - BEL_j;$$

SA_j = The sum assured (net of reinsurance) on sickness or disability for each policy where the benefits are payable as a single lump sum;

BA_i = The annualized amount of benefit (net of reinsurance) payable on sickness or disability for each policy j where the benefits are not payable as a single lump sum;

- 3.32 In calculating the capital requirement for catastrophe risk, the insurer may assume that management actions under stress scenarios can take place including changes to future bonus rates on policies with discretionary participation features. For life catastrophe risk, insurers should assume that the stress scenarios result entirely from industry-wide events.

Retrenchment risk

- 3.33 Retrenchment risk is the risk of loss or adverse changes in the value of insurance obligations resulting from changes in the level, trend or volatility of retrenchment inception rates.

- 3.34 The capital requirement for retrenchment risk, LU_{ret} shall be calculated according to the formula:

$$LU_{ret} = 60\% \times q \times n \sum_i \max\{CAR_i, 0\}$$

CAR_i = The capital-at-risk for product type i calculated as:

- i. The amount the insurer would currently pay should the persons insured under the policy get retrenched; plus

- ii. The expected present values of amounts not covered in (i) the insurer would pay should there be retrenchment of the persons insured under the policy; less
- iii. The best estimate liabilities of the corresponding obligations in points (i) and (ii);

(Each of the amounts above should be calculated net of amounts recoverable from eligible risk mitigation instruments.)

q = The insurer-specific expected average rate of retrenchment over the coming 12 months, weighted by the sum assured; and

n = The modified duration of the liability cash flows.

3.35 In calculating the capital requirement for retrenchment risk, the insurer may assume that management actions under stress scenarios can take place including changes to future bonus rates on policies with discretionary participation features.

3.36 The type and extent of management actions that the insurer may assume in the stress must consider whether the stress is insurer-specific or industrywide.

4. TS 6.2 A1: Glossary

Basic own funds (BOF): the excess of assets over liabilities, valued in accordance with ZICARP rules, adding back subordinated liabilities, and less any exclusions listed.

Capital-at-Risk (CaR): refers to the capital amount that is earmarked to cater for risks.

Catastrophe risk: The risk that a single event, or series of events, of major magnitude, usually over a short period (often 72 hours), leads to a significant deviation in actual claims from the total expected claims.

Confidence level: Statistical measure of the level of certainty regarding an outcome, typically expressed as the probability value $(1 - \alpha)$ associated with α confidence interval.

Expense risk: The risk of a change in value caused by the fact that the timing and/or the amount of expenses incurred differs from those expected, e.g., assumed for pricing basis.

Funeral insurance: A life policy with a low sum assured intended to pay for the burial costs on the death of the insured.

Group insurance: Contracts in which insurance cover is provided to a number of insured people (normally a workforce) or a number of other individual risks of one party, usually the counter-party, or many affected parties, which are not necessarily parties under the contract, under a single master contract.

Inflation risk: The risk of a change in value caused by a deviation of the actual market-consistent value of assets and/or liabilities from their expected value, due to inflation, e.g., price inflation, wage inflation, etc., leading to an unanticipated change in insurance cost and/or impact of an insurance contract, e.g. with respect to contract limits.

Insurance group: A group structure which contains two or more insurers.

Insurance contract: A contract under which one party (the insurer) accepts significant insurance risk from another party (the policyholder) by agreeing to compensate the policyholder or its beneficiary if a specified uncertain future event (the insured event) affects the policyholder.

Interest rate risk: The risk of a change in value caused by a deviation of the actual interest rates from the expected interest rates.

Lapse risk: The risks of a change in value caused by deviations from the actual rate of policy lapses from their expected rates.

Longevity risk: Type of biometric risk. A change in value caused by the actual mortality rate being lower than the one expected.

Management risk: Type of business risk. The risk associated with an incompetent management or a management with criminal intentions.

Market discipline: The creation of disciplining pressure through the publication of financial information and other information about the insurer's activities to the public, sometimes only to policyholders, providing transparency, hence allowing market participants and policyholders to assess key organizational and product information.

Market value: The amount for which an asset could be exchanged or a liability settled, between knowledgeable, willing parties in an arm's length transaction; based on observable prices within an active, deep and liquid market which is available to and generally used by the entity.

Morbidity risk: Type of biometric risk. A change of value caused by the actual disability and illness rates of the persons insured deviating from the ones expected.

Mortality risk: Type of biometric risk. A change in value caused by the actual mortality rate being higher than the one expected.

Reinsurance: Type of risk mitigation on the basis of an insurance contract between one insurer or pure reinsurer (the reinsurer) and another insurer or pure insurer (the cedant), to indemnify against losses, partially or fully, on one or more contracts issued by the cedant in exchange for a consideration (the premium).

Required economic capital: The total of assets measured at market-consistent value, internally required by an insurer above the market consistent value of obligations, in order to reduce the risk of not meeting the obligations to a defined risk measure (e.g., VaR, TVaR, EPD), and within a defined time period (e.g., one year).

Reserve risk: Relates to incurred claims, i.e., existing claims, (e.g., including IBNR and IBNER), and originates from claim sizes being greater than expected, differences in timing of claims payments from expected, and differences in claims frequency from those expected.

Risk margin: A generic term, representing the value of the deviation risk of the actual outcome compared with the best estimate, expressed in terms of a defined risk measure.

Solvency capital requirement: The amount of capital to be held by an insurer to meet the Pillar I requirements under ZICARP.

Technical provision: An insurer's contractual obligations/rights under an insurance contract.

Value-at-Risk: The maximum dollar amount expected to be lost (given normal market conditions) over a given time horizon, at a pre-defined confidence level. For example, if the 98% one-year VaR is \$1 million, there is a 99.5% confidence that over the next year, the portfolio will not lose more than \$1 million.



5. TS 6.2 A2 – Guidance on Selection of Mortality Tables

- 5.1 Insurance companies computing liabilities or estimating solvency capital under ZICARP may use the following mortality tables:
- i. SA56-62; or
 - ii. SA72-77; or
 - iii. SA85-90; or
 - iv. Any other later mortality tables, including those currently under development for Zimbabwe.
- 5.2 The choice of the final mortality table used to value technical provisions or calculate life underwriting should be dependent on the internal mortality experience of the insurer.
- 5.3 In the event that an insurer decides to use SA tables, the following minimum adjustments should be made:
- i. Internal mortality experience of the insurer;
 - ii. HIV/AIDS loadings; and
 - iii. Any other loadings that reflect differences in mortality experience between SA insureds and Zimbabwean insureds.